Name: _____

Class: _____Date:_____

# LESSON 12. COMMON MULTIPLES. LEAST COMMON MULTIPLE

2.44. Write the set of multiples of 6, the set of multiples of 9 and the set of common multiples of 6 and 9.

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2.45. Fill in the blanks with suitable words:

a) If 24 ⋮ a and 24 ⋮ b, then 24 is a ..... of a and b;

b) If 36 is the smallest non-zero natural number such that 36 ⋮ a and 36 ⋮ b, then 36 is the ..... of a and b.

2.46. Find the LCM of the two numbers m and n, given:

a) $m = 2^2 \cdot 5 \cdot 7^3$; $n = 2 \cdot 3^2 \cdot 5^3$;

b) $m = 3^4 \cdot 5 \cdot 11$; $n = 2 \cdot 3^2 \cdot 5^2 \cdot 7$.

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2.47. Find LCM(66, 110), then find the common multiples of 66 and 110.

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2.48. Find the LCM of:

a) 17 and 85;

b) 18; 45 and 90.

2.49. Three students on a field trip send text messages to tell their parents where they are. Nam sends a message every 40 minutes, Hà every 60 minutes and Mai every 90 minutes. If all three sent a message at 8 a.m., at what time will they next send messages at the same time?

2.50. Between 400 and 500 people took part in a group exercise practice. When the coach arranged them in rows of 6, rows of 9 or rows of 10, there were always 2 people left over. Exactly how many people took part?

2.51. Find the natural numbers a and b (a < b), given:

a) GCD(a, b) = 12 and LCM(a, b) = 180;

b) GCD(a, b) = 9 and LCM(a, b) = 252.

2.52. Write the following fractions with a common denominator:

a) $\frac{7}{18}$ and $\frac{5}{24}$;

b) $\frac{3}{4}$; $\frac{5}{6}$ and $\frac{7}{9}$.

2.53. In which year was the laptop computer invented? The first laptop was made in the year $\overline{abcd}$, where $\overline{abcd}$ is between 1950 and 1980 and is divisible by both 5 and 79. Find the year.

2.54. The Temple of Literature (Văn Miếu) in Hà Nội was built in the year $\overline{abcd}$ in the 11th century. Knowing that $\overline{abcd}$ is a four-digit number divisible by 2, 5 and 107, find the year it was built.

2.55. Part of a machine has two gears that fit together. Gear I has 30 teeth and gear II has 18 teeth. Two teeth that are touching each other are marked with "x". At least how many teeth must each gear turn so that the two marked teeth touch again in the same position? How many turns has each gear made then?